Project Title **:** Smart Crop Recommendation System using Machine Learning

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ABSTRACT

Agriculture is highly dependent on environmental conditions such as soil nutrients, rainfall, temperature, and humidity. Selecting the right crop is crucial for better yield and profit. This project proposes a Smart Crop Recommendation System using Machine Learning techniques to suggest suitable crops based on soil and weather conditions. The system analyzes parameters like N, P, K values, temperature, humidity, rainfall, and location data to predict the best crop. This helps farmers make informed decisions and improves agricultural productivity.

INTRODUCTION

Farmers often face difficulties in selecting the most suitable crop due to unpredictable weather and lack of soil analysis knowledge. Wrong crop selection leads to low yield and financial loss. To solve this problem, this project introduces a machine learning-based system that recommends the best crop based on environmental and soil conditions. The system uses data-driven techniques to assist farmers in making better agricultural decisions.

OBJECTIVES

- To recommend suitable crops based on soil and weather conditions
- To improve agricultural productivity using AI
- To analyze NPK values, temperature, rainfall, and humidity
- To support farmers in decision-making
- To reduce crop failure and financial loss

METHODOLOGY

- Collect dataset containing soil and weather parameters
- Preprocess and clean the dataset
- Perform feature selection (N, P, K, temperature, etc.)
- Train machine learning models
- Test and evaluate model accuracy
- Deploy model in web application

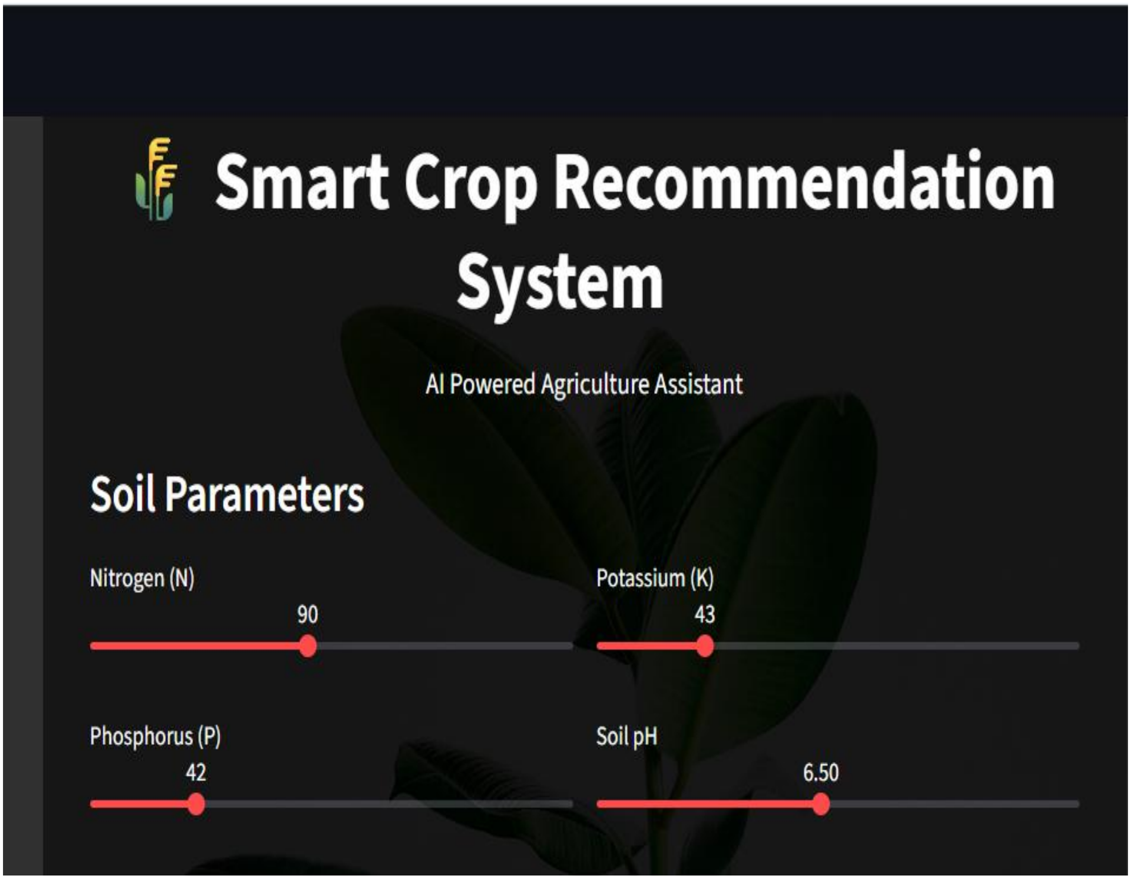
Technologies Used:

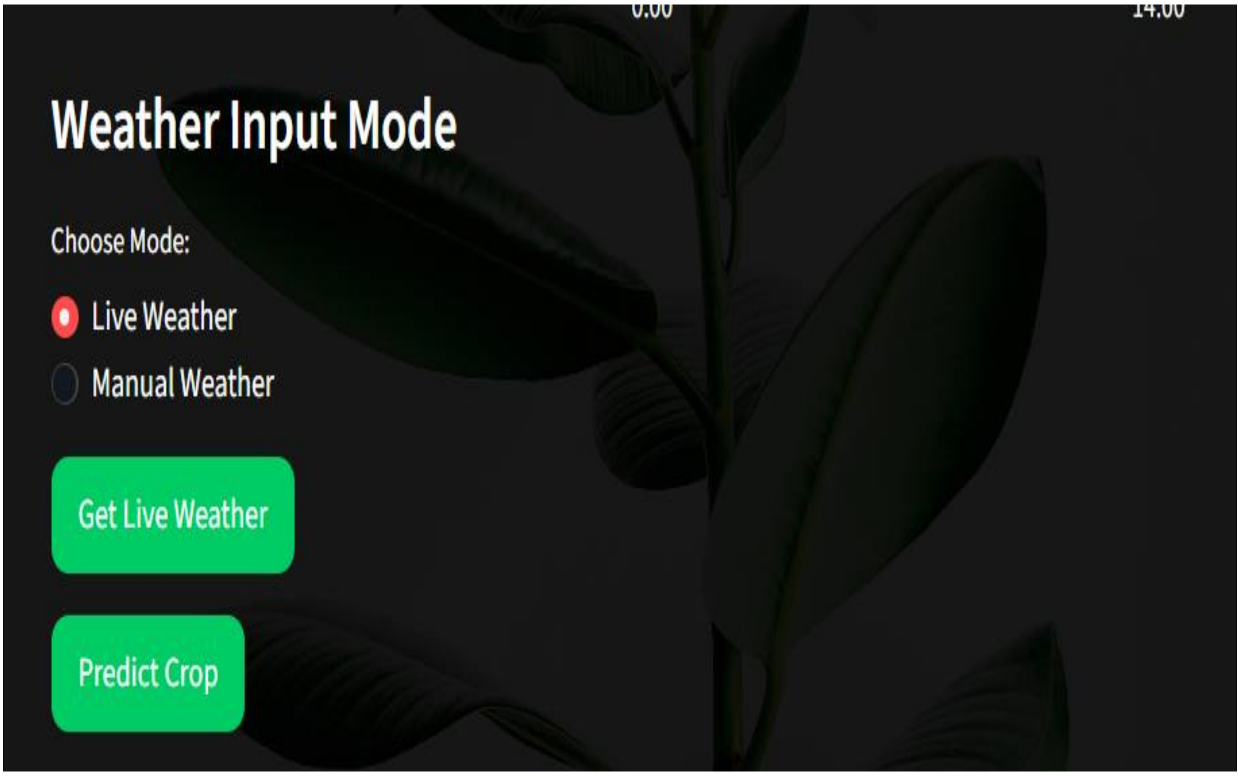
- Python
- Machine Learning (Scikit-learn)
- Pandas, NumPy
- Flask / Streamlit
- Weather API (for live data)

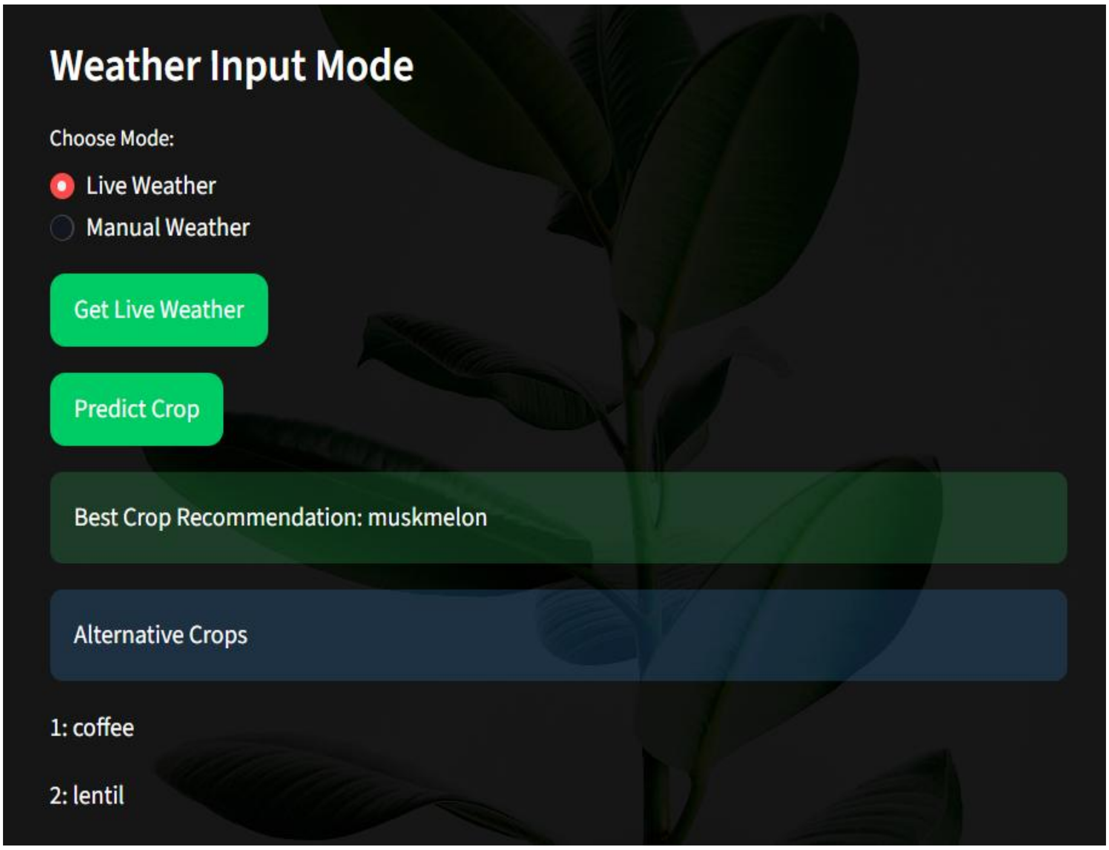
MODULES

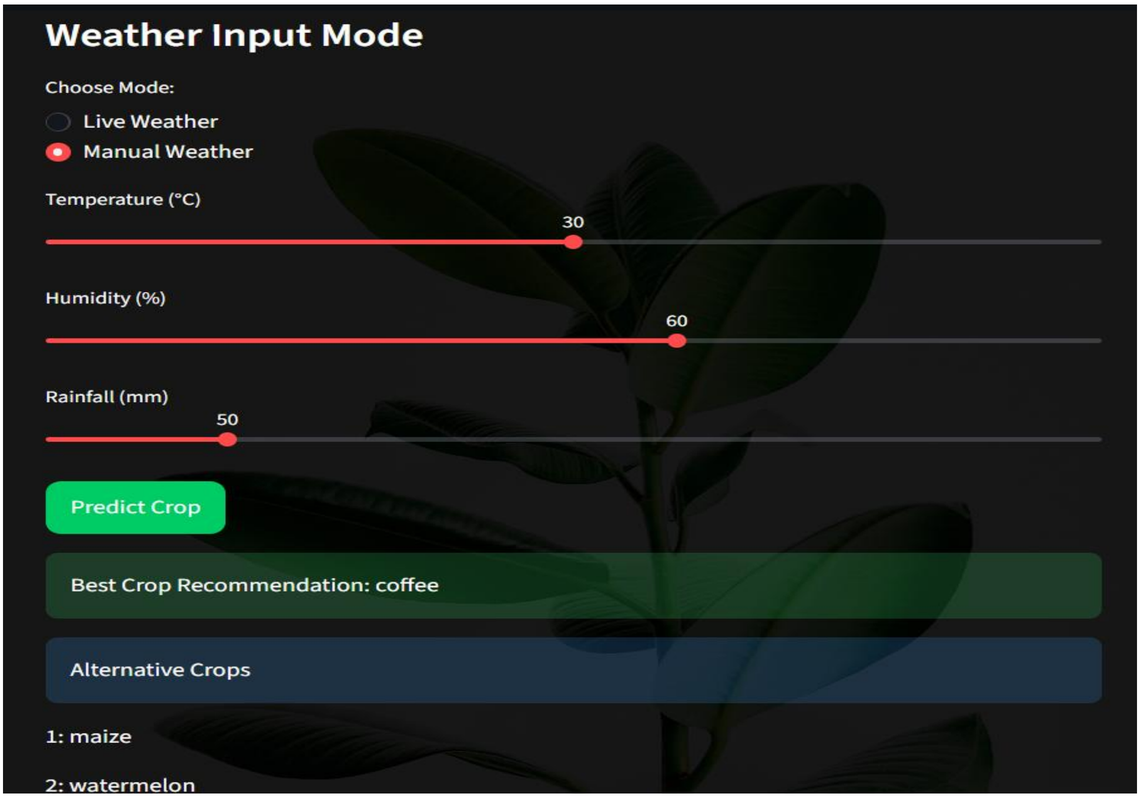
- Data Collection Module
- Data Preprocessing Module
- Soil Analysis Module
- Weather Data Integration Module
- ML Prediction Module
- Crop Recommendation Module
- Web Interface Module

OUTPUT SCREENSHOTS




0.00
14.00
Weather Input Mode
Choose Mode:
Live Weather
Manual Weather
Get Live Weather
Predict Crop


Weather Input Mode
Choose Mode:
Live Weather
Manual Weather
Get Live Weather
Predict Crop
Best Crop Recommendation: muskmelon
Alternative Crops
1: coffee
2: lentil



CONCLUSION

The Smart Crop Recommendation System successfully predicts the most suitable crop based on environmental and soil conditions using Machine Learning. It helps farmers make better decisions, reduces crop failure, and increases productivity. The system provides an efficient and intelligent solution for modern agriculture.

FUTURE SCOPE

- Fertilizer recommendation system
- Crop disease detection using images
- Mobile application development
- Integration with IoT sensors for real-time data
- Crop yield prediction system